Gemma 4 12B outperforms Qwen3.5-9B and Sailor2-8B on a native Filipino and Taglish evaluation, and tokenizes Filipino most efficiently: a pre-registered comparison

PinoAI research note · RESULT-001 · H-DEC-001

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! Correction (2026-09-30)

After publication, a scoring bug was found: the step that strips a leaked "thought" marker from one model's answers missed two of its four forms. With the fix, and the same registered method, two Gemma 4 12B answers change from wrong to right. **The verdict is unchanged, and the margin is wider.** Gemma vs Qwen: 36:10 ⟶ **36:8** ($p \approx 2.5 \times 10^{-5}$). Gemma vs Sailor2: 79:3 ⟶ **79:1**. The auto-scored-only sensitivity (S2): 9:2 ($p = 0.065$) ⟶ **9:0 ($p = 0.0039$)**, so the statement below that the automatically scored subset alone does not separate Gemma from Qwen **no longer holds**. The text and tables below are the original analysis, kept for the record.

Truncation (2026-09-30). The cut-off counts below (20, 12 and 24 of 175) came from a text-length proxy that undercounted badly. Measured with each model's own tokenizer, answers at the 512-token cap are 53 (Gemma 4 12B), 64 (Qwen3.5-9B) and 59 (Sailor2-8B) of 175, so the sentence "Gemma won despite more truncation than Qwen" below is **wrong**: Qwen was cut off more often. An exploratory, post-hoc check restricted to items where neither answer hit the cap still favours Gemma: 17:6 against Qwen ($p = 0.035$) and 32:1 against Sailor2 ($p < 10^{-8}$). Only 6 of Gemma's 36 wins over Qwen are items where only Qwen's answer was cut off. Truncation does not explain the result.

Abstract

We tested a hypothesis registered before any result: among three open-weight candidates for a Filipino foundation model, Gemma 4 12B (a) scores higher on PH-EVAL, a native-authored Filipino and Taglish evaluation, and (b) has the lowest Filipino token fertility.

Part A: PH-EVAL.

- Design: 175 items, the instruct versions of each model, thinking disabled, and a paired exact McNemar test that had to favour Gemma against both rivals.
- Result: Gemma 4 12B beat Qwen3.5-9B on discordant items 36 to 10 ($p = 0.0002$) and Sailor2-8B 79 to 3 ($p < 0.0001$).
- Accuracy: 78.3% for Gemma 4 12B, 63.4% for Qwen3.5-9B and 34.9% for Sailor2-8B.

Part B: tokenization. On 1,012 aligned FLORES+ sentences, Gemma 4's tokenizer used 1.840 tokens per Filipino word, against 1.974 for Qwen3.5 and 2.098 for Sailor2.

Both parts are supported. The result survives removing the ratings from a second review sheet whose blinding was compromised. It does not survive restricting the test to automatically scored items alone, where the Gemma-versus-Qwen difference is not significant ($p = 0.065$). This result supports, but does not by itself make, the base-model decision: a compute-feasibility gate and a base-checkpoint comparison come next.

1. Introduction

PinoAI adapts an open-weight model to Filipino and Taglish. Choosing the base model is the most consequential early decision, and it is easy to rationalise after the fact. We therefore registered the hypothesis, the test and the decision rule before running anything, and froze every later clarification before the first result.

2. Pre-registered hypothesis

H-DEC-001, stated by the project lead on 2026-09-28. It was recorded in the decision log at **2026-09-29T04:20Z** and committed to version control on its own before any Part A result:

> Gemma 4 12B scores higher than the other PH-Core candidates on PH-EVAL Filipino plus Taglish, and has lower Filipino token fertility.

Frozen before any result:

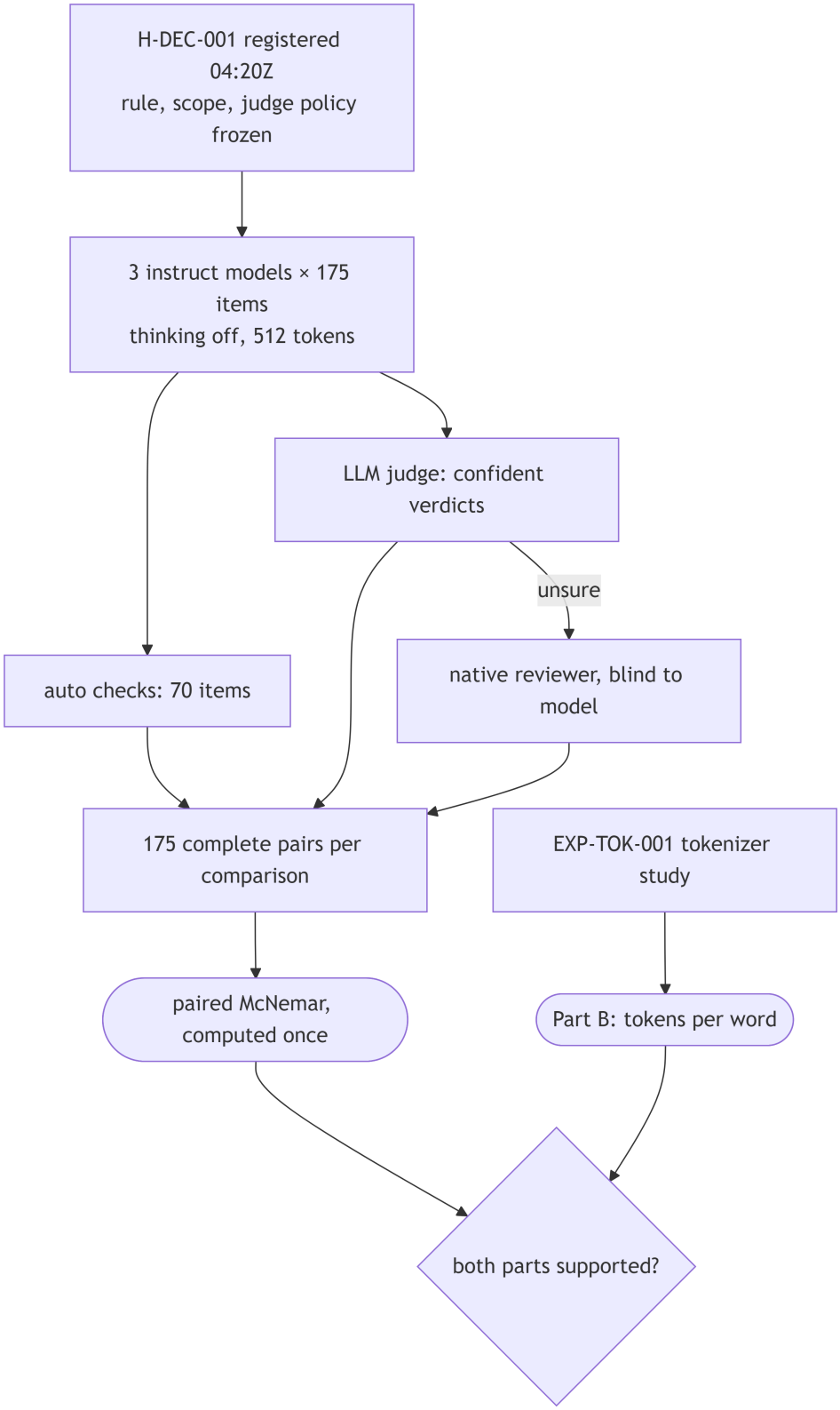
- **Part A test:** paired exact McNemar per pair, two-sided, $\alpha = 0.05$. Supported only if Gemma has more discordant wins and $p < 0.05$ against **both** rivals.
- **Pending items:** an item counts only if it is scored for both models in a pair; pending items are dropped, never imputed.
- **Scope:** all non-hidden PH-EVAL v0.1 items in Filipino or Taglish (175). This was confirmed by the project lead at 07:20Z.
- **Generation:** instruct versions, thinking disabled, 512 new tokens.
- **Judge policy:** the automatic judge policy in force at freezing, with human verdicts taking precedence. A later, retuned judge was explicitly **rejected** for this test, because it had been tuned on these outputs.
- **Part B:** Filipino tokens per word on FLORES+ `fil_Latn` devtest. Lower is better.

3. Method

- **Models (instruct versions, pinned revisions):**
  - `google/gemma-4-12B-it` rev. 707f0a3b8a
  - `Qwen/Qwen3.5-9B` rev. c202236235
  - `sail/Sailor2-8B-Chat` rev. 577d5b2745

  All ran in fp16 on Kaggle T4 GPUs with transformers 5.17.0. GPU time was 90.8, 83.7 and 60.7 minutes (driver wall clock).
- **Items:** PH-EVAL v0.1: 175 non-hidden items (105 Filipino, 70 Taglish). They are native-authored or reviewed, and cover reasoning, instruction following, structured output, safety and Taglish free generation.
- **Scoring, in order of precedence:**
  1. **Human:** a native-speaker reviewer rated every item the judge escalated, without model identity. In total, 236 rows were rated across two sheets.
  2. **Automatic:** exact-match and format checks for 70 items.
  3. **Judge:** confident verdicts of an LLM judge (judge policy v0.3); unsure cases escalated.

  After rating, 0 items were pending: 175 complete pairs for every comparison.
- **Statistics:** exact two-sided McNemar on discordant pairs, and 95% Wilson intervals for accuracy. The analysis ran once, on code written and reviewed before any result (`paired.py`), and a second team member recomputed the p-values.


H-DEC-001 registered
04:20Z
rule, scope, judge policy
frozen
3 instruct models × 175
items
thinking off, 512 tokens
LLM judge: confident
verdicts
unsure
auto checks: 70 items
native reviewer, blind to
model
175 complete pairs per
comparison
EXP-TOK-001 tokenizer
study
paired McNemar,
computed once
Part B: tokens per word
both parts supported?

4. Results

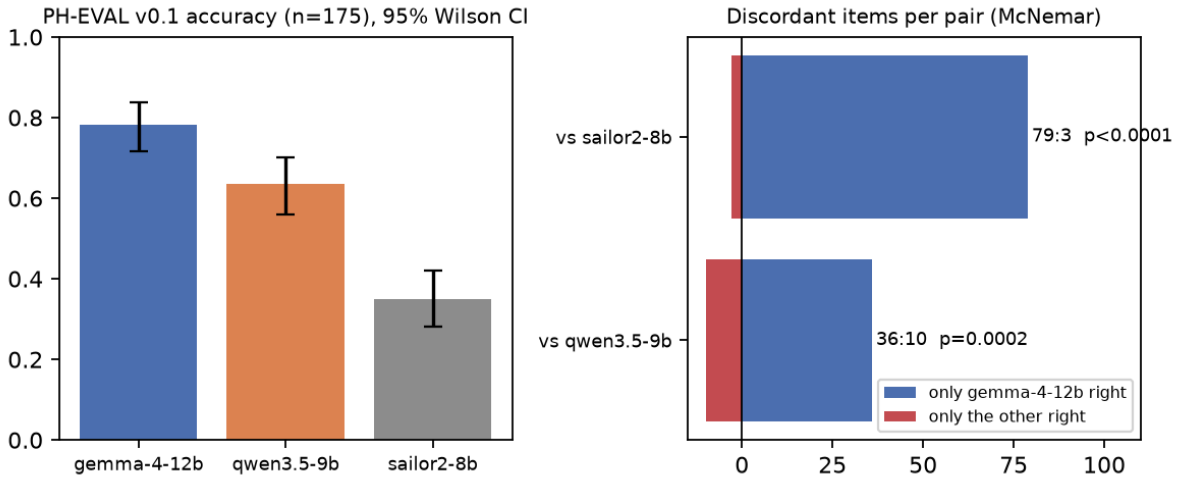


Figure 1: Left: PH-EVAL v0.1 accuracy with 95% Wilson intervals (n = 175 per model). Right: discordant items per pair; blue = only Gemma 4 12B correct, red = only the rival correct.

Table 1: Part A, primary analysis (pre-registered). Source: `experiments/results/DEC-001-partA.txt`.

pair	n	wins A:B	both right	both wrong	McNemar p	rule favours
gemma-4-12b vs qwen3.5-9b	175	36:10	101	28	0.0002	gemma-4-12b
gemma-4-12b vs sailor2-8b	175	79:3	58	35	< 0.0001	gemma-4-12b

Part A is supported. Gemma 4 12B is favoured in both pairs.

- **Accuracy:** 137/175 = 78.3% (Wilson 71.6–83.8%), against 111/175 = 63.4% for Qwen3.5-9B and 61/175 = 34.9% for Sailor2-8B.

Part B is supported. Previously reported and independently reproduced: 1.840 < 1.974 < 2.098 Filipino tokens per word (Gemma 4 < Qwen3.5 < Sailor2) [see the EXP-TOK-001 note].

H-DEC-001 is therefore supported.

Sensitivity analyses. These are reported, not substituted for the primary result.

Table 2: S1: the second review sheet's ratings removed (its answer key was exposed during rating); those items revert to pending and drop.

pair	n	wins A:B	both right	both wrong	McNemar p	rule favours
gemma-4-12b vs qwen3.5-9b	104	16:2	67	19	0.0013	gemma-4-12b
gemma-4-12b vs sailor2-8b	108	30:2	53	23	< 0.0001	gemma-4-12b

Table 3: S2: automatically scored items only (no judge, no human).

pair	n	wins A:B	both right	both wrong	McNemar p	rule favours
gemma-4-12b vs qwen3.5-9b	70	9:2	52	7	0.0654	no significant difference
gemma-4-12b vs sailor2-8b	70	20:2	41	7	0.0001	gemma-4-12b

Descriptive (not part of the test):

- **Qwen3.5-9B vs Sailor2-8B:** 59:9 (p < 0.0001).
- **Human-rated Taglish naturalness (0–3):** Gemma 4 12B 2.30 (n = 56), Qwen3.5-9B 1.96 (n = 57), Sailor2-8B 1.10 (n = 49).

- **Median Tagalog function-word share:** 0.97–1.00 for all three. None drifted to English. That contrasts with the earlier 7B model in our Taglish note (FINDING-001).

5. Discussion

The main result is robust in one direction and fragile in another:

- **It does not depend on the compromised sheet** (S1).
- **It does depend on human rating.** On the 70 automatically scored items alone, PH-EVAL v0.1 cannot separate Gemma 4 12B from Qwen3.5-9B (9:2, p = 0.065). Most of the evidence comes from free-generation items that only a human could score reliably. The judge escalated 66–89 of 105 per model.

That matches our pre-stated concern that v0.1 is underpowered for automatic scoring. It motivates PH-EVAL v0.2 with at least 500 automatically scorable items.

Sailor2′s weakness is notable. It is the only candidate whose pretraining explicitly targets Tagalog and Cebuano. Yet it has the lowest accuracy, the least natural Taglish and the least efficient tokenizer. Regional pretraining on an older base did not outweigh a newer general model.

Decision relevance. The registered reading order for the base-model decision puts a compute-feasibility gate first: a fixed A100-hour budget for a 2B-token first continued pretraining. Our pre-measurement estimate places Gemma 4 12B near or above that budget. If it fails the gate, the registered fallback is Gemma 4 E4B, which shares Gemma 4 12B's tokenizer, so Part B carries over. A base-checkpoint comparison (Part C) is also pending. This note therefore reports a supported hypothesis, not yet a chosen base.

6. Limitations and threats to validity

- **Blinding and rating speed (addressed, see §8).** During the second review sheet (164 rows), the answer key was accidentally uploaded into the reviewer's working session, and that sheet was rated at about 4 seconds per row. An independent second rater later agreed with the first on 25 of 26 of those rows (Cohen's $\kappa = 0.917$), and on 48 of 50 overlap rows overall ($\kappa = 0.913$). This clears the pre-set bar ($\kappa \geq 0.6$).
- **Two raters, one shared rubric.** The rubric was derived from the first rater's calibration notes. The agreement therefore measures consistency under a shared rubric, not fully independent judgment.
- **Rating design.** In the second sheet, each prompt appeared three times (once per model, shuffled). That allows relative judgments. It is acceptable for blind paired tests, but it is a property of the design.
- **Instruct proxies.** Part A tests instruct versions, including vendor post-training that we will not inherit. The base-checkpoint comparison (Part C) addresses this.
- **Settings.** Thinking was disabled for all models, so Part A measures non-thinking chat ability. Some answers hit the 512-token cap: 20, 12 and 24 of 175. Gemma won despite more truncation than Qwen.
- **Scale.** PH-EVAL v0.1 is small, with one Filipino/Taglish population and no Cebuano, Ilocano or Hiligaynon items.

7. Next steps

- Gate F: measure continued-pretraining throughput on the A100 for all three bases and Gemma 4 E4B, then apply the registered reading order.
- Part C: FilBench log-likelihood multiple choice on the base checkpoints.
- A second-rater validation, including the second sheet's rows.
- PH-EVAL v0.2: at least 500 automatically scorable items, plus Filipino math.

8. Update, 2026-09-30: independent second rater and judge validation

Two validation studies were registered before the fact, with their pass bars fixed in advance.

1. Inter-rater agreement. An independent second native reviewer (R2) rated a stratified 50-item overlap set blind to model identity and to the first reviewer's (R1) verdicts.

- **Pass/fail:** agreement 48/50, Cohen's $\kappa = 0.913$ (bar ≥ 0.6: **pass**).
- **By sheet:** on the 26 rows from the sheet whose key had been exposed, 25/26 ($\kappa = 0.917$). On the other rows, 23/24 ($\kappa = 0.909$).
- **Taglish naturalness (0–3):** quadratic-weighted $\kappa = 0.789$ ($n = 24$).

2. Validation of the judge's confident FAIL verdicts. Against human ratings, 6 of 40 confident fails were wrong: the humans passed the answer. The exact one-sided 95% upper bound is 27.5%, above the pre-set 15% bar, so the validation **fails**. The misses were privacy and instruction-following items.

Registered consequence: sensitivity S3. Part A was recomputed with every judge confident-fail treated as pending (dropped per pair). The primary result is not replaced.

Table 4: S3: Part A with the judge's confident-fail rows treated as pending. Source: `DEC-001-partA-S3-jevfails-pending.txt`.

pair	n	wins A:B	both right	both wrong	dropped	McNemar p	rule favours
gemma-4-12b vs qwen3.5-9b	145	26:5	101	13	30	0.0002	gemma-4-12b
gemma-4-12b vs sailor2-8b	134	61:2	58	13	41	< 0.0001	gemma-4-12b

Part A remains supported. Three of the four wrong fails in the Part A rows fell on Gemma 4 12B's answers, so the primary analysis was, if anything, conservative for Gemma.

Acknowledgments

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AI assistance. Analysis code, statistical checks and drafts of this document were prepared with the help of Claude (Anthropic), an AI model. Every number is generated from the project's result files. The author reviewed the content and takes responsibility for it.

References

Evaluation context: L. J. V. Miranda and others [1]; J. R. Montalan and others [2]. Background on the continued-pretraining plan: A. Grattafiori and others [3]; A. Ibrahim and others [4].

[1] L. J. V. Miranda and others, "FilBench: Can LLMs Understand and Generate Filipino?," *arXiv preprint arXiv:2508.03523*, 2025.

[2] J. R. Montalan and others, "Batayan: A Filipino NLP benchmark for evaluating Large Language Models," *arXiv preprint arXiv:2502.14911*, 2025.

[3] A. Grattafiori and others, "The Llama 3 Herd of Models," *arXiv preprint arXiv:2407.21783*, 2024.

[4] A. Ibrahim and others, "Simple and Scalable Strategies to Continually Pre-train Large Language Models," *arXiv preprint arXiv:2403.08763*, 2024.